BIO411/BIO629 2nd Mid-semester Exam (2025) IISER Mohali

Date: *11/10/2025* **Duration:** *60 min*

Maximum marks: *20*

Instructions:

a. *All questions in section A are compulsory. You can choose any three questions from Section B.*
b. *Answer questions briefly in section B. Include equations, wherever applicable.*

Section A

1. Find the sequence identity, positives and alignment score for the given alignment using BLOSUM62, affine gap penalty. (Gap opening and extension penalty of -10 and -1 respectively). (2 marks)

```
R Q P - - - S T V I
Q Q S L M M T T I L
```

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Score= 1+5-1-10-1-1+1+5+3+2
Identity: (2/10); Similarity: (6/10); alignment score:4
```

2. In a PSI-BLAST run, for a given query, the h-value in the search parameters is defined as inclusion threshold e-value cutoff to include sequences from hits in **the** PSSM construction for the next iteration (1 marks)
3. HSPs in BLAST refers to high-scoring segment pairs whose score cannot be increased by extending or reducing the alignment and E-value or expectation value is given by the equation = $K m n e^{-\lambda S}$ (1 marks)
4. In PAM100, 100 refers to 100 accepted point mutations per 100 amino acids. In BLOSUM45, 45 refers to percent identity threshold used for clustering sequences, i.e. sequence with >45% are clustered to construct substitution matrix. (1 marks)

Section B (Each question is of 5 marks)

5. Mr. X is studying the evolution of protein responsible for odor. For this study, homologues of odorant binding protein from fruit fly (3 proteins) and human (10 proteins) are collected. What is term coined to define relationships among proteins of fruit fly/humans? Which term is used to define relationship between fruit fly and human proteins? Propose a method to study with appropriate metric to determine if gene(s) is under evolutionary pressure to maintain its function.
   A. Paralogs, orthologs,
   B. To study if a gene is under evolutionary pressure to maintain its function, we will analyze selection acting on its coding sequence among orthologues. We will compare the rate of non-synonymous substitutions (**dN**) to the rate of synonymous substitutions (**dS**) or dN/dS ratio (one can use *Nei-Gojobori method for computing dN, and dS*). If dN/dS ratio is:

dN/dS= 1: Neutral evolution; **dN/dS < 1:** negative selection and protein function is under selective constraint or conserved in evolution; dN/dS **> 1:** Positive selection. Protein function evolution for adaptation.

6. Structures of bacterial truncated hemoglobin(A) and plant leghemoglobin (B) show remarkable structural similarity indicating strong possibility that sequences A and B share a common ancestor. On contrary, sequence identity between A and B is only 12%, suggesting otherwise. Propose a method to find whether sequences A and B are homologous to each other. Provide details of the method with main steps and reason why this method can be useful.

   To find whether A and B are homologous to each other, one needs to perform Position specific Iterated BLAST (PSI-BLAST). The reason PSI-BLAST can detect remote homologous relationship between sequences of low sequence identity “twilight zone” due to search performed by sensitive profile (PSSM), which embeds sequence position dependent substitution score. PSSM can capture position specific conservation and the same is reflected in profile, unlike BLOSUM62 matrix, which uses scoring constructed from all aligned positions. Steps: a. Query sequence search in sequence NR database, (BLAST), b. Using h-value cut-off collect sequences to perform MSA and PSSM is constructed, c. PSSM is used subsequent iterationss to find homologous sequences. d. Iterate until convergence.

7. Genome sequencing is followed by prediction of ORFs to build repository of all protein coding genes encoded in a genome. CpG island is also used in prediction of genes. Make the transition state diagram. Using a Hidden Markov model to predict CpG island (probability is given at the end) (No need to perform complete calculations):
   a. Find the most likely hidden path generating the sequence ‘ACG’. Check the correctness of the output.

   Use Viterbi algorithm to find most likely path to generate the sequence ‘AGC’.

State (R), sequence (C)	**A**	**G**	**C**
g (genome)	0.5x0.4=0.2	max[**0.95x0.2**,0.02x0.05]x0.2 =0.038 (gg)	max[**0.95x0.038**,0.02x0.0196]x0.2 =0.00722 (ggg)
i (island)	0.5x0.1=0.05	max[**0.98x0.05**,0.05x0.2]x0.4 =0.0196 (ii)	max[**0.98x0.0196**,0.05x0.0.038]x0.4 =0.00768 (iii)

Path is *iii* starting traceback from maximum of a score of 0.00768

Checking:

P(AGC|iii) = (0.5*0.1)x(0.98*.4)x(0.98*.4)=0.00768

   b. Which method can be used to find the probability of the sequence ‘ACG’. Suggest main step in calculating the probability.

P(AGC) will be found using forward algorithm.

State (R), sequence (C)	**A**	**G**	**C**
g (genome)	0.5x0.4=0.2	[0.95x0.2+0.02x0.05]x0.2 =0.0382	[0.95x0.0382+0.02x0.0236]x0.2 =0.007352
i (island)	0.5x0.1=0.05	[0.98x0.05+0.05x0.2]x0.4 =0.0236	[0.98x0.0236+0.05x0.0.0382]x0.4 =0.010015

P(AGC) = 0.007352+0.010015
= .017367

8. Which method will you use to determine sequence similarity between two sequences when they have large difference in their lengths? Justify your answer. What are linear and affine gap penalties? Compute local sequence alignment between two given sequences S1='HWPQRE' and S2='APQRWHE' using Match score =+3, Mismatch score = -1 and Gap =-2. Find alternate alignment by considering a score lower than maximum score.

The suggested sequence alignment method is to use local alignment because the global alignment while trying to align entire end-to-end sequence may introduce gaps (large penalties) and may miss relevant subsequence.

Linear gap penalty: $\gamma(g) = d \times n_g$ (d is gap penalty and n_g is length/no of gaps)

Affine gap penalty: $\gamma(g) = g_o + g_e(n_g-1)$; g_o and g_e are gap opening and extension penalty respectively. M=+3, MM=-1, G=-2

		A	**P**	**Q**	**R**	**W**	**H**	**E**
	0	0	0	0	0	0	0	0
H	0	0	0	0	0	0	3 (D)	1(L)
W	0	0	0	0	0	3 (D)	1 (LU)	2 (D)
P	0	0	3 (D)	1 (L)	0	1 (U)	2 (D)	0
Q	0	0	1 (U)	6 (D)	4 (L)	2 (L)	0	1 (D)
R	0	0	0	4 (U)	9 (D)	7 (L)	5 (L)	3 (L)
E	0	0	0	2 (U)	7 (U)	8 (D)	6 (DL)	8 (D)

Alignment starting from max score (9):

```
S1:PQR
S2:PQR

Score=8
S1:PQRW
S2:PQRE

Score=8
S1:PQRWHE
S2:PQR--E
```

Transition Probability	Genome	island
Genome	0.95	0.05
Island	0.02	0.98

Transition probability from start state to Genome/Island is 0.5

Emission Probability	Genome	island
A	0.4	0.1
C	0.2	0.4
G	0.2	0.4
T	0.2	0.1

BLOSUM62

A	4																			
R	-1	5																		
N	-2	0	6																	
D	-2	-2	1	6																
C	0	-3	-3	-3	9															
Q	-1	1	0	0	-3	5														
E	-1	0	0	2	-4	2	5													
G	0	-2	0	-1	-3	-2	-2	6												
H	-2	0	1	-1	-3	0	0	-2	8											
I	-1	-3	-3	-3	-1	-3	-3	-4	-3	4										
L	-1	-2	-3	-4	-1	-2	-3	-4	-3	2	4									
K	-1	2	0	-1	-1	1	1	-2	-1	-3	-2	5								
M	-1	-2	-2	-3	-1	0	-2	-3	-2	1	2	-1	5							
F	-2	-3	-3	-3	-2	-3	-3	-3	-1	0	0	-3	0	6						
P	-1	-2	-2	-1	-3	-1	-1	-2	-2	-3	-3	-1	-2	-4	7					
S	1	-1	1	0	-1	0	0	0	-1	-2	-2	0	-1	-2	-1	4				
T	0	-1	0	-1	-1	-1	-1	-2	-2	-1	-1	-1	-1	-2	-1	1	5			
W	-3	-3	-4	-4	-2	-2	-3	-2	-2	-3	-2	-3	-1	1	-4	-3	-2	11		
Y	-2	-2	-2	-3	-2	-1	-2	-3	2	-1	-1	-2	-1	3	-3	-2	-2	2	7	
V	0	-3	-3	-3	-1	-2	-2	-3	-3	3	1	-2	1	-1	-2	-2	0	-3	-1	4
	A	R	N	D	C	Q	E	G	H	I	L	K	M	F	P	S	T	W	Y	V